

Revised Load Interconnection System Impact Study Report

For Queue Project Q19-1349

Orebank 230 kV
February 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1349. The Interconnection Customer proposes a 176.8 MW large load (data center) facility interconnecting at the Orebank 230 kV substation in the Ohio area. The request was received on December 05, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 485 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,800,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of Dickinson to Ave Maria S 100kV L, the Marcos to Galeville 100kV line loads to 158.6% of its 163 MVA rating. Project DFAX: 5.0%.

The Winger to 20 Plant 100kV is overloaded to 144.5% of its 156 MVA emergency rating for the loss of the Imperial to Eufaula 500kV. The project contributes 9.3% DFAX (16.4 MW) to this overload.

88Th-Fruitridge to Hathaway 100kV: 113.1% loading (105 MVA) following the outage of the 20 Plant to Winger 100kV; DFAX = 6.9%.

For the loss of Imperial to Eufaula 500kV Line, the Nottingham to 20 Plant 100kV loads to 112.9% of its 113 MVA rating. Project DFAX: 6.2%.

For the loss of Imperial to Eufaula 500kV Line, the Juncos Tc to Jarvis 230kV L loads to 108.8% of its 491 MVA rating. Project DFAX: 9.1%.

Juniper to Chasm 100kV Line: 102.4% loading (134 MVA) following the outage of the Walteria 500/230 kV XFMR; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Marcos to Galeville 100kV line	Rebuild line with higher-capacity conductor	\$5,097,000	19% (\$990,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Winger to 20 Plant 100kV	Rebuild line with higher-capacity conductor	\$1,455,000	34% (\$487,000)
Hathaway to 88Th-Fruitridge 100kV	Rebuild line with higher-capacity conductor	\$2,175,000	27% (\$579,000)
Nottingham to 20 Plant 100kV	Rebuild line with higher-capacity conductor	\$2,213,000	34% (\$741,000)
Juncos Tc to Jarvis 230kV L	Reconductor line	\$3,452,000	51% (\$1,770,000)
Juniper to Chasm 100kV	Reconductor line	\$1,015,000	23% (\$233,000)
	Total allocated to project		\$4,800,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.